Bill of Materials - HMI Subsystem

Part Name/Description	Unit Quantity	Unit Prototype Cost	Total Prototype Cost	Unit Production Cost	Total Production Cost	Manufacturer	Manufacturer Part #	Vendor Link	Datasheet Link	Supplier	Supplier Part #	# Ordered	Date Ordered	# Received	Surplus	Schematic Reference Designators
ESP32-S3-WROOM-1 Module, SMD, 4MB Flash	1	$0.00	$0.00	$0.00	$0.00	Espressif	ESP32-S3-WROOM-1-N	https://www.digikey.com/en/products/detail/e		Peralta Labs	1965-ESP32-S3-WROOM-1-N4CT-N	0		0	-1	E1
Buck Switching Regulator, 3.3V Fixed, 3A, TO-263-5	1	$0.00	$0.00	$0.00	$0.00	Texas Instruments	LM2596S-3.3/NOPB	https://www.digikey.com/en/products/detail/t		Peralta Labs	LM2575T-3.3GOS-ND	0		0	-1	U1
USB Micro B Receptacle, 5-Pin, SMT Right Angle/THT	1	$0.00	$0.00	$0.00	$0.00	Amphenol	UX60SC-MB-5S8	https://www.digikey.com/en/products/detail/a		Peralta Labs	670-2675-1-ND	0		0	-1	J1
Connector Header, Through Hole, 8 Position, 2.54mm	5	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	61300811121	https://www.digikey.com/en/products/detail/v		Peralta Labs	609-5123-ND	0		0	-5	J3, J4, J5, J6, J7
DC Barrel Jack with Switch, 2.1mm ID, PCB Mount	1	$1.10	$1.10	$1.10	$1.10	CUI Devices	PJ-002BH	https://www.digikey.com/en/products/detail/c		DigiKey	CP-002BH-ND	0		0	-1	J2
0.96" I2C OLED Display Module, 128x64, SSD1306	1	$0.00	$0.00	$0.00	$0.00	Songhe	B085WCRS7C	https://www.amazon.com/dp/B085WCRS7C		Class	-	0		0	-1	LCD1
Schottky Rectifier Diode, 30V 5A, DO-201AD (1N5824 sub)	1	$0.62	$0.62	$0.62	$0.62	Vishay	SB530-E3/54	https://www.digikey.com/en/products/detail/v		DigiKey	SB530-E3/54GICT-ND	0		0	-1	D1
ESD Protection TVS Diode, 5V, SOD-923 (x2)	2	$0.17	$0.34	$0.17	$0.34	onsemi	ESD9B5.0ST5G	https://www.digikey.com/en/products/detail/onsemi/		DigiKey	ESD9B5.0ST5GOSCT-ND	0		0	-2	D6, D7
LED Blue, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051BS04000	https://www.digikey.com/en/products/detail/wurth-el		Peralta Labs	732-5013-ND	0		0	-1	D2
LED Green, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051VS04000	https://www.digikey.com/en/products/detail/wurth-el		Peralta Labs	732-5015-ND	0		0	-1	D3
LED White, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051WS04000	https://www.digikey.com/en/products/detail/wurth-el		Peralta Labs	732-5016-ND	0		0	-1	D4
LED Red, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051RS04000	https://www.digikey.com/en/products/detail/wurth-el		Peralta Labs	732-5011-ND	0		0	-1	D5
Inductor, 33uH, 2.05A, Radial (Bourns RLB9012)	1	$1.25	$1.25	$1.25	$1.25	Bourns	RLB9012-330KL	https://www.digikey.com/en/products/detail/bourns-		DigiKey	RLB9012-330KL-ND	0		0	-1	L1
Fuse, 1A Fast-Blow, 5x20mm Cylindrical	1	$0.45	$0.45	$0.45	$0.45	Littelfuse	0215001.MXP	https://www.digikey.com/en/products/detail/littelfuse		DigiKey	F1001-ND	0		0	-1	F1
Fuse Holder, 5x20mm, PCB Mount	1	$0.94	$0.94	$0.94	$0.94	Keystone	3544-2	https://www.digikey.com/en/products/detail/keyston		DigiKey	36-3544-2-ND	0		0	-1	F1 holder
Electrolytic Capacitor, 100uF 16V, Radial	1	$0.20	$0.20	$0.20	$0.20	Panasonic	ECA-1CM101	https://www.digikey.com/en/products/detail/panasor		DigiKey	P5138-ND	0		0	-1	C1
Electrolytic Capacitor, 220uF 16V, Radial	1	$0.25	$0.25	$0.25	$0.25	Panasonic	ECA-1CM221	https://www.digikey.com/en/products/detail/panasor		DigiKey	P5155-ND	0		0	-1	C2
Ceramic Capacitor, 0.1uF 50V, Through Hole (x3)	3	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-3	C3, C4, C5
Resistor, 10k, 2.5mW, Through Hole (x7)	7	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-7	R1, R2, R3, R4, R5, R6, R7
Resistor, 10k, Through Hole (x3)	3	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-3	R10, R11, R16
Resistor, 27 Ohm, Through Hole (x2)	2	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-2	R8, R9
Resistor, 470 Ohm, Through Hole (x2)	2	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-2	R12, R13
Resistor, 100 Ohm, Through Hole (x4) — LED current limiting	4	$0.00	$0.00	$0.00	$0.00	Various	Generic			Peralta Labs	Lab Reels	0		0	-4	R14, R15, R17, R18
Tactile Push Button Switch, SPST, Through Hole (x10)	10	$0.18	$0.00	$0.18	$0.00	E-Switch	TL1105SPF160Q	https://www.digikey.com/en/products/detail/e-switch		DigiKey	EG2506-ND	0		0	-4	SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10
2-Pin Jumper, 2.54mm (x3)	3	$0.05	$1.80	$0.05	$1.80	Wurth Elektronik	61000211121	https://www.digikey.com/en/products/detail/wurth-el		DigiKey	732-2860-ND	0		0	-10	JP1, JP2, JP3
Schottky Diode, 30V 200mA, SOD-323F (BAT54J)	1	0.18	0	0.18	0	Nexperia	BAT54J,115	https://www.digikey.co	https://assets.nexperia.	DigiKey	568-BAT54J,115CT-ND				0	D8
Fuse, 500mA Fast-Blow, 5x20mm Glass Cartridge	1	0.42	0.18	0.42	0.18	Littelfuse	0217.500MXP	https://www.digikey.co	https://www.littelfuse.c	DigiKey	F4744-ND				-1	F2
Test Point, Small PCB (x30)	30	0	0	0	0					Peralta Labs	-	0		0	-30	TP1-TP30
TOTALS			**$7.13**		**$7.13**											